# Inferring Latent Class Statistics from Text for Robust Visual Few-Shot Learning

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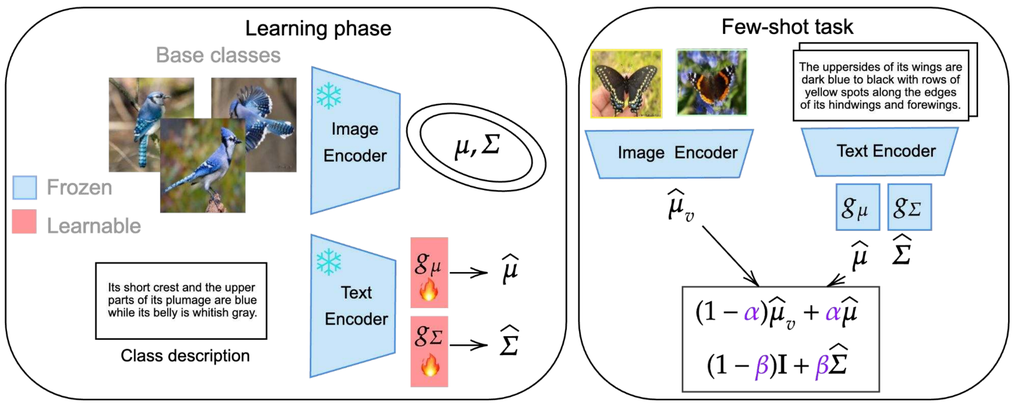
## Problem statement

- Multimodal foundation models like CLIP extract representations that preserve similarity
  - Text embeddings serve as class prototypes
- Q1: Do **text** embeddings yield information about higher moments of the **visual** embedding distribution?
- Q2: How much does text-inferred covariance improve generalization in **few-shot** tasks?

## TL;DR

- We train two mapping networks to **predict mean and covariance** of visual class distributions from text
- We show covariance predicted from text can bring additional information useful for **few-shot classification**

## Methodology




Using a base set, we train two shallow networks to learn the mapping from text to class mean and covariance. Then, given a few-shot task, we adapt the prediction of the mean using the available shots and apply shrinkage for the covariance matrix (assumed diagonal). We use the Mahalanobis distance for classification.

## Q1: How well do we predict mean and covariance from text?

We report MSE scores for both mean and covariance in various settings:

- In-domain (ID) / Cross-domain (CD)
- Class labels / descriptions
- iNaturalist / ImageNet1k

| Base dataset | | Model | $\mu$ val | $\mu$ test | $\Sigma$ val | $\Sigma$ test |
|---|---|---|---|---|---|---|
| iNaturalist | ID | Baseline | $1.00 \pm 0.02$ | $0.97 \pm 0.02$ | $1.00 \pm 0.03$ | $0.99 \pm 0.03$ |
| | | Class labels | $0.60 \pm 0.02$ | $0.60 \pm 0.02$ | $0.66 \pm 0.03$ | $0.72 \pm 0.03$ |
| | | Class description | $\mathbf{0.42 \pm 0.01}$ | $\mathbf{0.42 \pm 0.01}$ | $\mathbf{0.54 \pm 0.02}$ | $\mathbf{0.56 \pm 0.02}$ |
| ImageNet | ID | Baseline | $1.00 \pm 0.02$ | $0.98 \pm 0.02$ | $1.01 \pm 0.05$ | $1.00 \pm 0.04$ |
| | | Class labels | $\mathbf{0.44 \pm 0.02}$ | $\mathbf{0.45 \pm 0.01}$ | $\mathbf{0.70 \pm 0.03}$ | $\mathbf{0.74 \pm 0.04}$ |
| | | Class description | $0.46 \pm 0.02$ | $0.47 \pm 0.01$ | $\mathbf{0.70 \pm 0.03}$ | $\mathbf{0.74 \pm 0.04}$ |
| | CD | Baseline | $0.99 \pm 0.03$ | $1.67 \pm 0.10$ | $1.04 \pm 0.07$ | $4.99 \pm 0.90$ |
| | | Class labels | $\mathbf{0.43 \pm 0.03}$ | $\mathbf{1.26 \pm 0.11}$ | $0.72 \pm 0.06$ | $\mathbf{3.99 \pm 0.60}$ |
| | | Class description | $0.45 \pm 0.02$ | $1.30 \pm 0.11$ | $\mathbf{0.70 \pm 0.05}$ | $4.67 \pm 0.84$ |

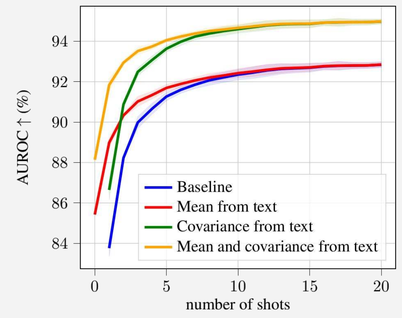
**MSE of predictions. The baseline is the average value over the training set**

## Q2: How much does text-inferred covariance improve generalization?

### One-class

$$p(x|\mu,\Sigma) = \frac{1}{\sqrt{(2\pi)^d|\Sigma|}} \exp\left(-\frac{1}{2}(x-\mu)^T\Sigma^{-1}(x-\mu)\right)$$

**In-domain (iNaturalist) 5-shot tasks**



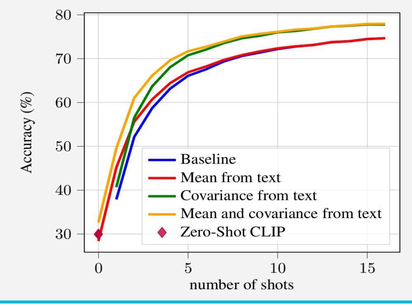

**Cross-domain**

| | Method | Shots | Caltech | EuroSAT | Food | Flowers | SUN397 | DTD | Pets | Cars | UCF101 | Average |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| AUROC | Baseline | 1 | 89.74 | 81.34 | 83.34 | 90.92 | 89.18 | 71.23 | 81.47 | 90.31 | 84.70 | 84.70 |
| | | 2 | 93.31 | 85.79 | 87.59 | 94.66 | 94.37 | 76.36 | 86.16 | 94.28 | 88.63 | 89.02 |
| | | 4 | 94.95 | 88.60 | 89.54 | 96.38 | 96.47 | 79.53 | 88.11 | 96.04 | 90.70 | 91.15 |
| | | 8 | 95.89 | 89.85 | 90.36 | 97.09 | 97.51 | 80.88 | 88.83 | 96.78 | 91.72 | 92.10 |
| | | 16 | 96.49 | 90.39 | 90.79 | 97.51 | 97.83 | 81.63 | 89.21 | 97.12 | 92.25 | 92.57 |
| Δ% | Mean from text (M) | 1 | +2.07 | 0.00 | +0.68 | +0.48 | +2.83 | +0.17 | +6.31 | +0.38 | +1.37 | 1.59 |
| | | 2 | +0.46 | 0.00 | 0.00 | 0.00 | +0.22 | 0.00 | +2.17 | 0.00 | +0.11 | 0.33 |
| | | 4 | +0.10 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | +0.76 | 0.00 | 0.00 | 0.10 |
| | | 8 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | +0.37 | 0.00 | 0.00 | 0.04 |
| | | 16 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | +0.18 | 0.00 | 0.00 | 0.02 |
| | Covariance from text (C) | 1 | +2.09 | +1.47 | +3.05 | +3.59 | +0.08 | 0.00 | +4.57 | +1.97 | 0.00 | +1.86 |
| | | 2 | +1.80 | +1.45 | +3.09 | +2.53 | +0.21 | 0.00 | +3.38 | +1.66 | 0.00 | +1.57 |
| | | 4 | **+1.49** | **+1.27** | **+2.92** | **+1.82** | **+0.21** | **0.00** | **+2.76** | **+1.23** | **0.00** | **+1.30** |
| | | 8 | **+1.15** | **+1.22** | **+2.73** | **+1.45** | **+0.22** | **0.00** | **+2.53** | **+1.07** | **0.00** | **+1.15** |
| | | 16 | **+0.96** | **+1.20** | **+2.64** | **+1.16** | **+0.25** | **0.00** | **+2.40** | **+1.00** | **0.00** | **+1.07** |
| | Mean and covariance from text (M&C) | 1 | **+3.82** | **+1.47** | **+3.61** | **+3.99** | **+2.85** | **+0.17** | **+8.43** | **+2.33** | **+1.37** | **+3.11** |
| | | 2 | **+2.07** | **+1.45** | **+3.09** | **+2.54** | **+0.34** | 0.00 | **+4.18** | **+1.66** | **+0.11** | **+1.71** |
| | | 4 | **+1.52** | **+1.27** | **+2.92** | **+1.82** | **+0.21** | 0.00 | **+2.79** | **+1.23** | +0.01 | **+1.31** |
| | | 8 | **+1.15** | **+1.22** | **+2.73** | **+1.45** | **+0.22** | 0.00 | **+2.53** | **+1.07** | 0.00 | **+1.15** |
| | | 16 | **+0.96** | **+1.20** | **+2.64** | **+1.16** | **+0.25** | 0.00 | **+2.40** | **+1.00** | 0.00 | **+1.07** |

### Multi-class

$$p(y=c|x) = \frac{\exp\left(\sqrt{(\hat{\mu}_c - \mathbf{x})^T \hat{\mathbf{\Sigma}}_c^{-1}(\hat{\mu}_c - \mathbf{x})}\right)}{\sum_{c'} \exp\left(\sqrt{(\hat{\mu}_{c'} - \mathbf{x})^T \hat{\mathbf{\Sigma}}_{c'}^{-1}(\hat{\mu}_{c'} - \mathbf{x})}\right)}$$

**In-domain (iNaturalist) 5-ways 5-shot tasks**




**Cross-domain**

| | Method | Shots | Caltech | EuroSAT | Food | Flowers | SUN397 | DTD | Pets | Cars | UCF101 | Average |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Zero-Shot CLIP | 0 | 82.39 | 36.01 | 75.72 | 62.28 | 56.60 | 40.01 | 79.59 | 51.91 | 59.42 | 60.43 |
| Accuracy | Baseline | 1 | 62.07 | 47.80 | 27.55 | 52.25 | 28.15 | 26.11 | 26.69 | 21.62 | 36.89 | 36.57 |
| | | 2 | 74.95 | 58.30 | 40.33 | 69.10 | 41.42 | 37.64 | 38.28 | 33.18 | 50.08 | 49.25 |
| | | 4 | 81.31 | 66.07 | 51.67 | 80.49 | 51.89 | 47.49 | 49.65 | 44.02 | 58.80 | 59.04 |
| | | 8 | 84.61 | 70.67 | 59.72 | 86.52 | 58.59 | 54.17 | 58.63 | 52.53 | 64.23 | 65.52 |
| | | 16 | 86.34 | 73.49 | 64.68 | 89.59 | 62.46 | 58.21 | 65.23 | 65.18 | 67.48 | 70.68 |
| Δ% | Mean from text (M) | 1 | +16.46 | +0.82 | +13.80 | +6.58 | +17.22 | +6.94 | +26.56 | +5.26 | +9.98 | +11.51 |
| | | 2 | +6.30 | +0.33 | +8.09 | +2.58 | +9.00 | +3.62 | +19.74 | +2.04 | +4.06 | +06.20 |
| | | 4 | +2.32 | -0.07 | +3.85 | +0.75 | +3.66 | +1.49 | +13.40 | +0.58 | +1.53 | +03.06 |
| | | 8 | +0.76 | -0.19 | +1.57 | +0.20 | +1.30 | +0.55 | +8.91 | +0.13 | +0.51 | +01.53 |
| | | 16 | +0.21 | -0.33 | +0.67 | +0.04 | +0.43 | +0.28 | +5.61 | +5.76 | +0.15 | +0.91 |
| | Covariance from text (C) | 1 | +1.56 | +0.84 | +4.06 | +6.34 | +0.74 | +1.79 | +6.64 | +4.93 | +1.48 | +03.15 |
| | | 2 | +2.92 | +1.52 | +6.72 | +6.84 | +2.16 | +3.75 | +11.10 | +7.58 | +3.50 | +05.12 |
| | | 4 | +2.85 | +1.97 | +7.41 | +5.35 | +3.21 | +5.27 | +13.29 | +8.72 | +4.55 | +05.85 |
| | | 8 | +2.44 | +2.03 | +6.71 | +3.88 | +3.59 | +5.44 | +12.60 | +8.77 | +4.98 | +05.60 |
| | | 16 | +2.29 | +1.99 | +5.80 | +2.95 | +3.59 | +5.33 | +11.31 | +11.22 | +5.11 | +04.82 |
| | Mean and covariance from text (M&C) | 1 | **+21.03** | **+2.20** | **+25.57** | **+14.90** | **+22.14** | **+14.05** | **+37.86** | **+16.14** | **+14.94** | **+18.76** |
| | | 2 | **+10.27** | **+2.04** | **+18.42** | **+9.82** | **+13.74** | **+10.45** | **+31.86** | **+11.22** | **+9.48** | **+13.03** |
| | | 4 | **+5.76** | **+2.21** | **+12.49** | **+6.21** | **+8.23** | **+8.15** | **+25.80** | **+9.67** | **+7.27** | **+09.53** |
| | | 8 | **+3.71** | **+2.09** | **+8.78** | **+4.21** | **+5.56** | **+6.71** | **+20.42** | **+9.06** | **+6.29** | **+07.43** |
| | | 16 | **+2.81** | **+2.02** | **+6.74** | **+3.13** | **+4.46** | **+6.10** | **+16.30** | **+15.99** | **+5.91** | **+5.94** |

### Setup

- Baseline: Nearest Class Mean classifier
- 1 000 runs for each setting
- Cross-domain base dataset: ImageNet-1k
- Covariance matrix assumed diagonal

### Discussion

- Using the text mean usually improves performance in extreme low-shot (1-2) setting
- Covariance brings more consistent gains across various shot regimes
- Using the mean and the covariance together brings the best gains
- Our methods outperform zero-shot CLIP in in-domain

## Future work

- Impact of hyper-parameters on the prediction (α, β…)
- Predicting higher order moments of the class visual distributions
- Applicability to other tasks:
  - Generative tasks: using the mean and covariance to guide sampling from the class conditional distributions
  - Semantic segmentation: prompting for Segment Anything using mean and covariance